

Statistics : It can be defined ^{as} the science which deals with methods of collecting, classifying, presenting, comparing and interpreting numerical data collected to reach to a conclusion & hence help in decision making.

Measures of Central Tendency :-

These are the measures that are used for describing the central location of a set of data values and are representing the entire data.

- Arithmetic Mean
- Median
- Mode
-

Mean: Mean, or average, is the sum of all values in a dataset divided by the number of values.

• It represents the central point around which the data is distributed.

• It helps in decision making.

• What is outlier in machine learning.

$\{2, 3, 4, 5, 8, 9\} = A$

$A = \{2, 3, 4, 5, 8, 9, 32\}$

↓
outlier.

Median: - Median is the middle value of a dataset when it is ordered.

Mode: - It is the value that appears the most frequently in a dataset.

eg Again, ^{help} in decision making.

eg if a New company wants to manufacture a product, then he will do survey of Mode of product.

Formula:

$$\text{Mean} = \bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

$$\text{Mean} = \bar{x} = \frac{1}{N} \sum_{i=1}^n f_i x_i, \quad \begin{matrix} x_1, x_2, \dots, x_n \\ f_1, f_2, \dots, f_n \end{matrix}, \quad N = \sum f_i$$

Teacher's Signature _____

Median - It is the value which divides the data set into two equal parts when arranged in increasing (or, decreasing) order. The median is thus a positional average.

$n = \text{odd}$

$$\text{Median} = \left(\frac{n+1}{2}\right)^{\text{th}} \text{ observation}$$

$n = \text{even}$

$$\text{Median} = \frac{\left(\frac{n}{2}\right)^{\text{th}} + \left(\frac{n}{2} + 1\right)^{\text{th}} \text{ observation}}{2}$$

Mean: Average: The total of the numbers divided by numbers

eg: $\{20, 10, 40, 30\}$

$$\begin{aligned}\text{Mean} &= \frac{20 + 10 + 40 + 30}{4} \\ &= \frac{100}{4} = 25\end{aligned}$$

eg $\{20, 10, 40, 30, 90\}$

$$\text{Mean} = \frac{190}{5} = 38$$

* For given data, 90 is outlier.

* Outliers affect data.

Median = $\{20, 10, 40, 30, 90\}$

Sort data = $\{10, 20, 30, 40, 90\}$

Median = 30

eg $A = \{10, 20, 30, 40, 90, 100\}$

$$\text{Median} = \frac{30 + 40}{2} = \frac{70}{2} = 35$$

Mode =

Set = $\{10, 20, 10, 30, 40, 10, 90\}$

No. frequency

10

3

20

1

30

1

40

1

90

1

$\therefore \text{Mode} = 10$

The following data gives the frequencies of serum cholesterol level of 1000 males aged between 25 to 35 years arrived at a particular city hospital during the last one year.

Cholesterol level	No. of males
80 - 120	12
120 - 160	145
160 - 200	380
200 - 240	292
240 - 280	118
280 - 320	35
320 - 360	11
360 - 400	07

Find (a) Mean (b) Median (c) Mode.

Class Interval	Mid value (x_i)	Frequency f	$u = \frac{x - 180}{40}$	fu
80 - 120	100	12	-2	-24
120 - 160	140	145	-1	-145
160 - 200	180	380	0	0
200 - 240	220	292	1	292
240 - 280	260	118	2	236
280 - 320	300	35	3	105
320 - 360	340	11	4	44
360 - 400	380	07	5	35
		<u>1000</u>		<u>591</u>

$$\text{Mean} = \bar{x} = a + h\bar{u} = 180 + 40 \left(\frac{591}{1000} \right) = 203.64$$

$$\text{where } \bar{u} = \frac{\sum fu}{\sum f}$$

Teacher's Signature _____

Median.	frequency (f)	c.f (cumulative frequency)
Class Interval		
80-120	12	12
120-160	145	157
160-200	380	537
200-240	292	829
240-280	118	947
280-320	35	982
320-360	11	993
360-400	7	1000
	1000	

$\frac{N}{2} = 500$ Thus 160-200 is the median class, $l = 160$, $f = 380$, $h = 40$, $c = 157$

$$\text{Median} = l + \frac{h}{f} \left(\frac{N}{2} - c \right) = 160 + \frac{40}{380} (500 - 157)$$

$$\text{Median} = 196.11$$

② Mode :-

∵ Maximum frequency is 380.

Thus 160-200 is the modal class.

$$l = 160, f_m = 380, f_1 = 145, f_2 = 292$$

$$h = 40$$

$$\text{Mode} = l + \frac{(f_m - f_1)h}{2f_m - (f_1 + f_2)} = 160 + \frac{(380 - 145)40}{2(380) - (145 + 292)}$$

$$\text{Mode} = 189.10$$

Variance : A measurement of how far each number in a data set is from the mean (average), & thus from every other number in the set.

Let us take two series.

$$S_1 = \{4, 6\} \Rightarrow \bar{x}_1 = \frac{4+6}{2} = \frac{10}{2} = 5$$

$$S_2 = \{1, 9\} \Rightarrow \bar{x}_2 = \frac{1+9}{2} = \frac{10}{2} = 5$$

In series S_1 , data points are close to $\bar{x}_1$ (Less variance)

In series S_2 , data points are far from $\bar{x}_2$ (More Variance)

→ For discrete data :

$$\text{Variance} = \sigma^2 = \frac{\sum (x_i - \mu)^2}{N} \rightarrow \text{Population Variance}$$

$$\text{For } S_1 = \frac{(4-5)^2 + (6-5)^2}{2} = 1 \quad \text{Less Variance}$$

$$\text{For } S_2 = \frac{(1-5)^2 + (9-5)^2}{2} = 16 \quad \begin{array}{l} \text{More Variance} \\ \text{More dispersion} \end{array}$$

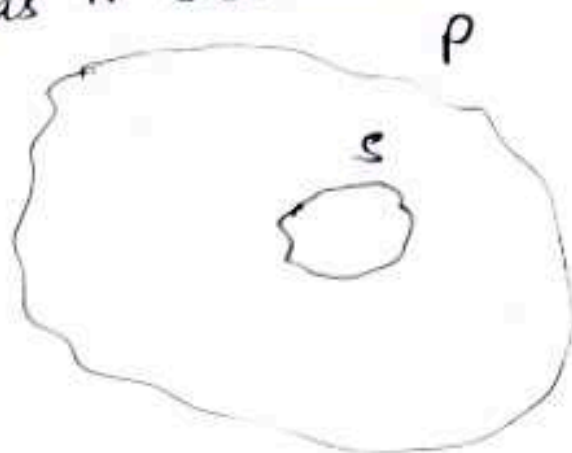
Sample Variance:

Let S be a sample of a population P .
 S has n -elements. P has N elements

then

$$\text{Sample Variance} = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{(n-1)}$$

Where $\bar{x}$ is Mean of Sample



Variance for continuous data

$$\sigma^2 = \frac{1}{N} \sum f(x - \bar{x})^2$$

$$\text{or } \sigma^2 = \left(\frac{\sum f d^2}{N} - \left(\frac{\sum f d}{N} \right)^2 \right) \times h^2$$

where $d = \frac{x - a}{h}$

Q Calculate variance of data:

Age	20-30	30-40	40-50	50-60	60-70	70-80	80-90	
No.	3	61	132	153	140	51	2	

y

Age	No. of members	x_i	$d = \frac{x_i - a}{h}$	fd	fd^2
20-30	3	25	-3	-9	27
30-40	61	35	-2	-122	244
40-50	132	45	-1	-132	132
50-60	153	55	0	0	0
60-70	140	65	1	140	140
70-80	51	75	2	102	204
80-90	2	85	3	6	18
$\Sigma f = 542$				$-15 = \Sigma fd$	$\Sigma fd^2 = 765$

$a = \text{Assumed Mean} = 55$

$$\mu = a + \frac{\Sigma fd}{\Sigma f} \cdot h = 55 + \frac{(-15) \times 10}{542}$$

$$\Rightarrow \mu = 54.72$$

$$\text{Variance} = \sigma^2 = \left(\frac{765}{542} - \left(\frac{-15}{542} \right)^2 \right) (10)^2$$

$$\sigma^2 = 141.07$$

- Random experiment: An experiment that can result in different outcomes, even though it is repeated in the same manner every time, is called a random experiment.
- Sample space: The set of all possible outcomes of a random experiment is called the sample space of the experiment. The sample space is denoted by S . For eg. If a fair coin is tossed two times then $S = \{HH, HT, TH, TT\}$
- Discrete & Continuous Sample space:

<u>Discrete sample space</u>	<u>Continuous sample space</u>
A sample space is discrete if it consists of a finite or countably infinite set of outcomes	A sample space is continuous if it contains an interval of real numbers
eg. Rolling a fair die = $\{1, 2, 3, 4, 5, 6\}$	eg. Observing the distance a cricket ball can be thrown d , $S = \{d: d \geq 0\} = \mathbb{R}^+$

Event: An event is a subset of the sample space of a random experiment.

eg: Experiment $\rightarrow$ Tossing a fair coin two times

$$S = \{HH, TT, HT, TH\}$$

event $A =$ getting two heads $\{HH\}$

$\rightarrow$ Basic Set of Operations:-

- **Union**:- It is the event that consist of all outcomes that are contained in either of two events, denoted as $E_1 \cup E_2$

- **Intersection**:- It is the event that consist of all outcomes that are contained in both of events. $E_1 \cap E_2$

- **Compliment**:- It is the set of outcomes in the sample space that are not in the event E^c or E'

Q If the experiment consist of flipping two coins, then

$$S = \{HH, HT, TH, TT\}$$

Then $E_1 =$ Event that at least one head appears =

$$\Rightarrow E_1 = \{(HH), (HT), (TH)\}$$

& $E_2 =$ Event that at least one tail appears

$$E_2 = \{(HT), (TH), (TT)\}$$

Here $E \cup F = S$, $E \cap F = \{(HT), (TH)\}$
 $E^c = \{(TT)\}$

→ Mutually Exclusive Events:- Two events E_1 & E_2 are Mutually Exclusive events if $E_1 \cap E_2 = \phi$

→ Probability: It is used to quantify the likelihood, or chance, that an outcome of a random experiment will occur.

→ Probability of an Event:- The probability of an event refers to the likelihood that the event will occur.

Let A be an event then $P(A)$ = Probability of event A = $\frac{\text{No. of Favourable Outcomes}}{\text{Total No. of outcomes}}$

→ Exhaustive Events:- A set of events is called Exhaustive if all the events together consumes the entire sample space.

eg Tossing a coin

sample space = $S = \{H, T\}$

$A = \text{getting head}$
 $B = \text{getting tail}$ } $\Rightarrow A \text{ \& } B \text{ makes sample space}$

→ Definition of Probability:- If there are n exhaustive, mutually exclusive & equally likely cases of which m are favorable to an event E , then $P(E) = \frac{m}{n} = \frac{\text{Favourable}}{\text{Total}}$

Principal of Counting:- If an event can happen in n_1 ways & therefore for each of these exist a second event can happen in n_2 ways, and for each of these first and second events, a third event can happen for n_3 ways & so on, then the number of ways these n events can happen is given by the product $n_1 \cdot n_2 \cdot \dots \cdot n_n$.

Permutations:- In permutation, the elements should be arranged in a particular order.

For eg: Set $A = \{1, 6\}$. Permutation of A is 2 i.e. $\{1, 6\}$ & $\{6, 1\}$

→ The no. of Permutations of n different elements is $n!$ Where $n! = n(n-1)(n-2) \dots 2 \cdot 1$.

→ Permutations of Subsets:- The no. of permutations of subsets of r -elements selected from a set of n -different elements is ${}_n P_r = \frac{n!}{(n-r)!} = n(n-1)(n-2) \dots (n-r+1)$

→ Permutations of Similar objects:- The no. of permutations of $n = n_1 + n_2 + \dots + n_k$ objects of which n_1 are of one type, n_2 are of second type, $\dots$ and n_k of k th type is $n!$

$$n_1! n_2! n_3! \dots n_k!$$

→ Combinations:- It is the selection of r objects taken

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out of n -objects irrespective of the objects arrangements.

$${}^nC_r = \binom{n}{r} = \frac{n!}{(n-r)! r!}$$

8 From six engineers and five architects a committee is to be formed having three engineers and two architects.

How many different committees can be formed if (i) there is no restriction (ii) two particular engineers must be included.

(iii) One particular architect must be excluded.

(i) No. of committee having 3 engineers & 2 architect = ${}^6C_3 \times {}^5C_2 = 200$

(ii) No. of committee = ${}^4C_1 \times {}^5C_2 = 40$

(iii) No. of committee = ${}^6C_3 \times {}^4C_2 = 120$

9 What is the chance that a leap year selected at random will contain 53 Sundays?

Soln No. of days in a leap year = 366 days
year = $(52 \times 7 + 2 \text{ days})$

∴ Leap year has 52 weeks & 2 extra days

These 2 extra days = ① Mon, Tue, ② Tue, Wed

③ Wed, Thurs ④ Thurs, Fri ⑤ Fri, Sat ⑥ Sat, Sun

⑦ Sun, Mon

∴ Probability of 53 Sundays = $\frac{2}{7}$

A Urn contains 5 red and 10 black

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balls. Eight of them are placed in another urn. What is the chance that the latter contains 2 red & 6 black balls.

The no. of ways in which 8 balls can be drawn out of 15 is ${}^{15}C_8$.

The no. of ways of drawing 2 red balls = 5C_2 .

The no. of ways of drawing 6 black balls = ${}^{10}C_6$.

∴ total no. of ways in which 2 red & 6 black balls can be drawn is ${}^5C_2 \times {}^{10}C_6$.

$$\therefore \text{required probability} = \frac{{}^5C_2 \times {}^{10}C_6}{{}^{15}C_8}$$
$$= \frac{140}{429}$$

→ Axioms of Probability :-

① For any event E of sample space S,
 $0 \leq P(E) \leq 1$

② The sum of probabilities of all sample events is unity or $\sum_{i=1}^n P(E_i) = 1$

③ $P(E_1 \cup E_2) = P(E_1) + P(E_2)$; $E_1 \cap E_2 = \phi$

Addition Rule :

- Mutually Exclusive : Those events which do not occur together

eg. Tossing a coin

As head and tail together can't come.

$$P(A \cup B) = P(A) + P(B)$$

- NOT Mutually Exclusive Events : events occur together

eg. getting a king of heart

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$\text{eg } P(\text{king of heart}) = \frac{4}{52} + \frac{13}{52} - \frac{1}{52} = 16/52$$

$$P(\text{heart}) = 13/52$$

$$P(\text{king}) = \frac{4}{52}$$

$$\text{eg } P(\text{Ace or king}) = \frac{4}{52} + \frac{4}{52} = 8/52$$

$$\text{eg } P(\text{heart or spade}) = \frac{13}{52} + \frac{13}{52} = 26/52$$

$$\text{eg } P(\text{Queen or Black}) = \frac{4}{52} + \frac{26}{52} - \frac{2}{52}$$

$$= \frac{28}{52}$$

OR $\rightarrow$ Addition
And $\rightarrow$ Multiplication

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$P(A \cup B)$

$P(A + B)$

$P(A \cup A)$

$P(\text{getting } A \text{ or } B)$ $P(A \text{ either } A \text{ or } B)$

$P(\text{at least one})$

$\rightarrow$ To find these we Apply
Addition Law

Multiplication Rule

Independent Events:

Occurring of one event does not
affect other event

$$P(A \cap B) = P(A) \times P(B)$$

Dependent Event

Occurring of one event affect
other event

$$P(A \cap B) = P(A) \times P(B/A)$$

2. Two coins are tossed what
is the probability of getting
head on first & tail on second.

$$P(H, T) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$$

Q Two packets of playing cards. What is the probability of getting Ace from first pack & king from second pack

$$P(A, K) = \frac{4}{52} \times \frac{4}{52}$$

Q There is one packet of playing cards. Two cards are drawn one by one without replacement. What is the probability of getting Ace on first draw & king on second draw.

(Dependent event)

$$P(A, K) = \frac{4}{52} \times \frac{4}{51}$$

$P(A \cap B)$

$P(A \cap B)$

$P(\text{both})$

$P(A \text{ as well as } B)$

Use Multiplication Rule

Conditional Probability :-

Let A & B be two events of a sample space S and let $P(B) \neq 0$. Then conditional probability of A , given B , denoted by $P(A/B)$, is defined as

$$P(A/B) = \frac{P(A \cap B)}{P(B)}$$

Bayes' Thm :- Let $B_1, B_2, \dots, B_n$ are mutually exclusive events with $P(B_i) \neq 0$ ($i=1, 2, \dots, n$) of a random experiment then for any arbitrary event A of sample space of above experiment with $P(A) > 0$

$$P(B_i/A) = \frac{P(B_i) P(A/B_i)}{\sum_{i=1}^n P(B_i) P(A/B_i)}$$

Ques A student knew only 60% of the questions in a test each with 5 answers. He simply guessed while answering the test. What is the probability that he knew the ans. to a question given that he answered it correctly.

Q1) A_1 : student knew the answer
 A_2 : student guess " "
 E : student answered it correctly

$$P(A_1) = 0.60$$

$$P(A_2) = 0.40$$

$$P(E|A_1) =$$

$$P(A_1|E) = \frac{P(E|A_1) \cdot P(A_1)}{P(E|A_1)P(A_1) + P(E|A_2)P(A_2)}$$

$$P(E|A_1) = 1$$

$$P(E|A_2) = 1/5$$

$$P(A_1|E) = \frac{1 \times 0.60}{1 \times 0.60 + \frac{1}{5} \times 0.40}$$

$$= \frac{0.60}{0.60 + 0.08} = \frac{0.60}{0.68}$$

$$= 0.88235$$

Q From a bag containing 3 white, 5 black balls, 4 ball balls are transferred into an empty bag. From this bag, a ball is drawn and is found to be white. What is the probability that out of four balls transferred

Q. A card from a pack of 52 cards is lost. From the remaining cards of the pack, two cards are drawn and found to be heart. Find the probability of the remaining cards to be a heart.

Let E_1 = missing card is a heart
 E_2 = " " " club
 E_3 = " " " diamond
 E_4 = " " " spade

A = Drawing two heart cards from remaining cards

$$P(E_1) = \frac{13}{52} = \frac{1}{4}$$

illy $P(E_2) = P(E_3) = P(E_4) = \frac{1}{4}$

$$P(A|E_1) = \frac{{}^{12}C_2}{{}^{51}C_2}$$

$$P(A|E_2) = \frac{{}^{13}C_2}{{}^{51}C_2}$$

$$P(A|E_4) = \frac{{}^{13}C_2}{{}^{51}C_2}$$

$$P(A|E_3) = \frac{{}^{13}C_2}{{}^{51}C_2}$$

$$P(E_1|A) = \frac{P(E_1) \cdot P(A|E_1)}{P(A|E_1)P(E_1) + P(A|E_2) \cdot P(E_2) + P(A|E_3)P(E_3) + P(A|E_4)P(E_4)}$$

$$P(E_1|A) = 11/50$$

Q An insurance company insured 3000 scooters, 4000 cars, 5000 trucks.

Prob. of accident due to scooter = 0.002

car = 0.03

truck = 0.04

One of the insured vehicles met with an accident. Find the prob. that it is a car.

Solⁿ Let E_1 : event of driving a scooter
 E_2 : " " " " car
 E_3 : " " " " truck
 A : event of occurrence of an accident.

$$P(E_2|A) = \frac{P(A|E_2) P(E_2)}{P(A|E_1) P(E_1) + P(A|E_2) P(E_2) + P(A|E_3) P(E_3)}$$

$$P(E_1) = \frac{3}{12} = \frac{1}{4}$$

$$P(E_2) = \frac{4}{12} = \frac{1}{3}$$

$$P(E_3) = \frac{5}{12}$$

$$= \frac{\frac{2}{100} \times \frac{4000}{12000}}{\frac{2}{100} \times \frac{1}{4} + \frac{3}{100} \times \frac{1}{3} + \frac{4}{100} \times \frac{5}{12}}$$

$$= \frac{\frac{2}{100} \times \frac{4000}{12000}}{\frac{2}{100} \times \frac{1}{4} + \frac{3}{100} \times \frac{1}{3} + \frac{4}{100} \times \frac{5}{12}}$$

$$\Rightarrow P(E_2|A) = \frac{6}{19}$$

①

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Random Variable : It is a real valued function which assigns a real number to each sample point in the sample space.

eg Tossing a fair coin twice

sample space $S = \{HH, TT, HT, TH\}$

X (Number of head) = Random variable

$$X(0) = 1$$

$$X(1) = 2$$

$$X(2) = 1$$

Discrete Random Variable : A random variable which takes finite or at most countable number of values is called Discrete Random Variable.

eg Experiment = Rolling a die
Sample space = $\{1, 2, 3, 4, 5, 6\}$

$$X(\text{No. of even outcomes}) = 3$$

$$X(\text{No. of odd outcomes}) = 3.$$

Discrete Probability Distribution : It is random variable with its corresponding probability.

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eg Tossing a fair coin twice

$$S = \{HH, TT, HT, TH\}$$

 X (No. of Tails) = Random Variable

$$X(0) = 1$$

$$P(X=0) = 1/4$$

$$X(1) = 2$$

$$P(X=2) = 2/4$$

$$X(2) = 1$$

$$P(X=1) = 1/4$$

X	0	1	2
$P(X)$	$1/4$	$2/4$	$1/4$

Q 3 white balls are mixed with 5 Red balls in a box. Find the probability distribution of No. of white balls if two balls are drawn

Soln

3 W
5 R

 $X \rightarrow$ No. of White balls

$$P(X=0) = \frac{{}^5C_2}{{}^8C_2}$$

$$P(X=1) = \frac{{}^5C_1 {}^3C_1}{{}^8C_2}$$

$$P(X=2) = \frac{{}^3C_2}{{}^8C_2}$$

X	0	1	2
$P(X)$	$\frac{{}^5C_2}{{}^8C_2}$	$\frac{{}^5C_1 {}^3C_1}{{}^8C_2}$	$\frac{{}^3C_2}{{}^8C_2}$

↓
Probability
Distribution

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Probability Mass Function ∴

Let X be a discrete random variable such that $P(X=x_i) = p_i$ then p_i is said to be probability mass function if it satisfy following condition ∴

$$(i) \quad p_i \geq 0 \quad i = 1, 2, 3 \dots n.$$

$$(ii) \quad \sum p_i = 1$$

Q Tossing a fair coin twice

X	0	1	2
$P(X)=p_i$	$1/4$	$2/4$	$1/4$

Here $p_i \geq 0$

$$\sum p_i = 1 = \frac{1}{4} + \frac{2}{4} + \frac{1}{4}$$

Hence p_i is Probability Mass fn

Distribution Function

Let X be a discrete Random variable then its discrete distribution function or cumulative distribution function (C.d.f) is defined as

$$F(x) = \sum p_i = P(X \leq x_i)$$

Q Tossing a fair coin twice.

$$f(x) = \begin{cases} 1/4 & x \leq 0 \\ 3/4 & x \leq 1 \\ 4/4 & x \leq 2 \end{cases}$$

Cumulative distribution fn

Q following is probability distribution

X	0	1	2	3	4	5	6	7
P(X)	0	K	2K	2K	3K	K ²	2K ²	7K ² +K

Find (i) K

(ii) $P(X < 6)$, $P(X \geq 6)$, $P(0 < X < 5)$

(iii) Distribution function

(iv) if $P(X \leq K_1) > 1/2$ find value of K_1

(v) Find $P\left(\frac{1.5 < X < 4.5}{X > 2}\right)$

(1) $\sum P_i = 1$

$$\Rightarrow 0 + K + 2K + 2K + 3K + K^2 + 2K^2 + 7K^2 + K$$

$$\Rightarrow 9K + 10K^2 = 1$$

$$\Rightarrow 10K^2 + 9K - 1 = 0$$

$$\Rightarrow (K+1)(10K-1) = 0$$

$$\Rightarrow K = -1, K = 1/10$$

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But $K = -1$ is not possible as probability of any event is not negative

$$\therefore K = \frac{1}{10}$$

(ii)

X	0	1	2	3	4	5	6	7
P(X)	0.4	0.1	0.2	0.2	0.3	0.01	0.02	0.17

(ii)

$$P(X < 6) = (0 + 0.1 + 0.2 + 0.2 + 0.3 + 0.01)$$

$$P(X < 6) = 0.81$$

OR

$$P(X < 6) = 1 - P(X \geq 6)$$

$$= 1 - [0.02 + 0.17]$$

$$P(X < 6) = 0.81$$

$$P(X \geq 6) = P(X=6) + P(X=7)$$

$$= 0.02 + 0.17$$

$$= 0.19$$

⑥

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C.d.f

$$f(x) = F(x) = \begin{cases} 0 & x \leq 0 \\ 0.1 & x \leq 1 \\ 0.3 & x \leq 2 \\ 0.5 & x \leq 3 \\ 0.8 & x \leq 4 \\ 0.81 & x \leq 5 \\ 0.83 & x \leq 6 \\ 1 & x \leq 7 \end{cases}$$

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$$P(X \leq 0) = 0$$

$$P(X \leq 1) = 0.1$$

$$P(X \leq 2) = 0.3$$

$$P(X \leq 3) = 0.5$$

$$P(X \leq 4) = 0.8 > \frac{1}{2}$$

$$\therefore K_1 = 4$$

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$$P\left(\frac{1.5 < X < 4.5}{X > 2}\right) = \frac{[P(1.5 < X < 4.5) \cap P(X > 2)]}{P[X > 2]}$$

$$= \frac{P(X=2, 3, 4) \cap P(X=2, 3, 4)}{1 - P(X \leq 2)}$$

$$= \frac{P(3) + P(4)}{1 - [P(0) + P(1) + P(2)]}$$

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$$= \frac{0.2 + 0.3}{1 - (0 + 0.1 + 0.2)} = 5/7$$

Continuous Random Variable :-

A random variable which can take infinite no. of values in an interval is known as continuous Random variable

eg Height of students in class 12th

eg Weight of a group of individuals

Probability Density function (P.D.f)

For a continuous R.V. X ,

A function $f(x)$ is p.d.f

if (i) $f(x) \geq 0$

(ii) $\int_{-\infty}^{\infty} f(x) dx = 1$

(iii) $P(a \leq X \leq b) = \int_a^b f(x) dx$

= area under $f(x)$ from a to b

NOTE: It is also known as density function.

⑧

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eg: If X is a continuous r.v. with following p.d.f.

$$f(x) = \begin{cases} \alpha(2x-x^2) & 0 < x < 2 \\ 0 & \text{else.} \end{cases}$$

① find α

② $P(X > 1)$

solⁿ ① By defⁿ of p.d.f.

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

$$\Rightarrow \int_{-\infty}^0 f(x) dx + \int_0^2 f(x) dx + \int_2^{\infty} f(x) dx = 1$$

$$\Rightarrow \int_0^2 \alpha(2x-x^2) dx = 1$$

$$\Rightarrow \alpha \left(\frac{x^2}{2} - \frac{x^3}{3} \right) \Big|_0^2 = 1$$

$$\Rightarrow \alpha \left(4 - \frac{8}{3} \right) = 1 \quad \boxed{\alpha = 3/4}$$

② $P(X > 1) = \int_1^{\infty} f(x) dx$

$$= \int_1^2 \frac{3}{4} (2x-x^2) dx$$

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$$\Rightarrow P(X > 1) = \frac{3}{4} \left(x^2 - \frac{x^3}{3} \right) \Big|_1^\infty$$

$$\Rightarrow P(X > 1) = \frac{3}{4} \left(\frac{2}{3} \right) = \frac{1}{2}$$

Q A random variable X has density function

$$f(x) = \begin{cases} Kx^2 & -3 < x < 3 \\ 0 & \text{else} \end{cases}$$

① find K

② $P(1 \leq X \leq 2)$, $P(X \leq 2)$, $P(X > 1)$

Solⁿ By defⁿ of P.d.f

$$\int_{-\infty}^{\infty} f(x) dx = 1$$

$$\Rightarrow \int_{-3}^3 Kx^2 dx = 1 \Rightarrow 2 \int_0^3 Kx^2 dx = 1$$

$$\Rightarrow 2K \left(\frac{x^3}{3} \right)_0^3 = 2K \left(\frac{27}{3} \right) = 1$$

$$\Rightarrow \boxed{K = 1/18}$$

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$$(11) P(1 \leq X \leq 2) = \int_1^2 f(x) dx$$

$$= \frac{1}{18} \int_1^2 x^2 dx = \frac{1}{18} \left[\frac{x^3}{3} \right]_1^2$$

$$= \frac{1}{18} \left(\frac{8}{3} - \frac{1}{3} \right) = 7/54$$

$$P(X \leq 2) = \int_{-\infty}^2 f(x) dx.$$

$$= \frac{1}{18} \int_{-3}^2 x^2 dx = \frac{1}{18} \left[\frac{x^3}{3} \right]_{-3}^2$$

$$P(X \leq 2) = \frac{1}{18} \left(\frac{8}{3} + \frac{27}{3} \right) = \frac{35}{54}$$

$$P(X > 1) = \int_1^{\infty} f(x) dx$$

$$= \frac{1}{18} \int_1^3 x^2 dx$$

$$= \frac{1}{18} \left(\frac{x^3}{3} \right)_1^3 = 26/54.$$

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Cumulative Cumulative distribution function / (CDF)

Continuous distribution function:

Let X be a continuous R.V. having p.d.f $f(x)$ then.

$F_X(x)$ will be continuous distribution function of X if

$$F_X(x) = P(X \leq x) = \int_{-\infty}^x f(x) dx.$$

NOTE : Relation between
Distribution function and
density function.

$$\boxed{\frac{d}{dx} F(x) = f(x)}$$

Q. The probability density function of R.V. X is

$$f(x) = \begin{cases} x & 0 \leq x < 1 \\ 2x & 1 \leq x < 2 \\ 0 & \text{else} \end{cases}$$

find (i) $P(X \geq 1.5)$
(ii) find c.d.f

NOTE: - If X is a continuous probability distribution then probability at a single point is always zero. (12)

$$(i) P(X \geq 1.5) = \int_{1.5}^{\infty} f(x) dx$$

$$= \int_{1.5}^2 (2-x) dx = \left(2x - \frac{x^2}{2} \right)_{1.5}^2$$

$$\Rightarrow P(X \geq 1.5) = 0.125$$

(ii) find c.d.f.

$$P(X \leq x) = \int_{-\infty}^x f(x) dx$$

$$0 < x < 1$$

$$f(x) = \int_{-\infty}^0 f(x) dx + \int_0^x f(x) dx$$

$$= 0 + \int_0^x x dx = x^2/2$$

$$x \leq 2$$

$$f(x) = \int_{-\infty}^0 f(x) dx + \int_0^1 f(x) dx + \int_1^x f(x) dx$$

$$= \int_0^1 x dx + \int_1^x (2-x) dx$$

$$= \left(\frac{x^2}{2} \right)_0^1 + \left(2x - \frac{x^2}{2} \right)_1^x$$

$$= 1/2 + 2x - \frac{x^2}{2} - \left(2 - \frac{1}{2} \right) = -1 + 2x - \frac{x^2}{2}$$

Q.72

$$f(x) = \int_{-\infty}^0 f(x) + \int_0^1 f(x) dx + \int_1^2 f(x) dx +$$

$$\int_2^{\infty} f(x) dx =$$

$$F_X(x) = \begin{cases} 0 & -\infty < x < 0 \\ x^2/2 & 0 < x \leq 1 \\ -1 + 2x - x^2/2 & 1 < x \leq 2 \\ 1 & x > 2 \end{cases}$$

↓

$$\frac{d}{dx} F_X(x) = f(x)$$

Mathematical Expectation

Various Measures of Central tendency

Let X be any Random variable and $\phi(x)$ be any function of x . Then expectation of $\phi(x)$ is denoted by $E[\phi(x)]$ & is defined as

Discrete R.V. $\downarrow$ $E[\phi(x)]$ continuous R.V. $\downarrow$

$$\sum \phi(x) p(x)$$

$$\int_{-\infty}^{\infty} \phi(x) f(x)$$

• Mean of X denoted by $E(X)$:-

$$E(X) = \sum_{i=1}^{\infty} x_i P(X=x_i) \rightarrow X \rightarrow \text{Discrete R.V.}$$

$$E(X) = \int_{-\infty}^{\infty} x f(x) dx \rightarrow X \rightarrow \text{Continuous R.V.}$$

$$\text{variance} = E[X - \bar{X}]^2 = E[X^2 - 2X\bar{X} + \bar{X}^2]$$

$$\begin{aligned} \text{variance}(X) &= E(X^2) - 2\bar{X} E(X) + \bar{X}^2 \\ &= E(X^2) - 2\bar{X}^2 + \bar{X}^2 \end{aligned}$$

$$= E(X^2) - \bar{X}^2$$

$$\boxed{\text{Var}(X) = E[X^2] - (E(X))^2}$$

$$\text{Var}(X) = \sigma^2 = \sum_{i=1}^n (x_i - \mu)^2 p_i$$

$$\text{mean}(X) = \mu = \sum_{i=1}^n x_i P(x_i)$$

Q Find mean and variance of Probability distribution given by the following table

x_i	1	2	3	4	5
$P(x_i)$	0.2	0.35	0.25	0.15	0.05

$$\underline{\underline{\text{Soln}}}$$

$$E(X) = \sum_{i=1}^n x_i P(x_i)$$

$$\begin{aligned} &= 1 \times 0.2 + 2 \times 0.35 + \\ &3 \times 0.25 + 4 \times 0.15 + 5 \times 0.05 \end{aligned}$$

$$\boxed{\text{Mean} = 2.5}$$

$$\begin{aligned} \text{Now } E[X^2] &= \sum x_i^2 p_i \\ &= (1^2 \times 0.2) + (2^2 \times 0.35) + \\ &\quad (3^2 \times 0.25) + (4^2 \times 0.15) + 5^2 \times 0.05 \end{aligned}$$

$$E(X^2) = 7.5$$

$$\begin{aligned} \text{var}(X) &= (7.5)^2 - (2.5)^2 \quad \left(\because \text{var}(X) = \right. \\ &\quad \left. E(X^2) - (EX)^2 \right) \\ &= 7.5 - 6.25 \end{aligned}$$

$$\boxed{\text{var}(X) = 1.25}$$

Q. Thirteen cards are drawn simultaneously from a pack of 52 cards. If ace, count 1, face count 10 & others according to the denomination. Find the expectation of total score in 13 cards.

$$\text{ex}^n \begin{array}{c|c|c|c|c|c|c|c|c|c|c} X & 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 & 9 \\ \hline p & \frac{1}{13} & \frac{1}{13} & \frac{1}{13} & \frac{1}{13} & \frac{1}{13} & \frac{1}{13} & \frac{1}{13} & \frac{1}{13} & \frac{1}{13} \end{array}$$

$$\begin{array}{c|c|c|c|c} X & 10 & 10 & 10 & 10 \\ \hline p_i & \frac{1}{13} & \frac{1}{13} & \frac{1}{13} & \frac{1}{13} \end{array}$$

$$E(X) = \frac{1}{13} (1+2+3+4+5+6+7+8+9+10+10+10+10)$$

$$\Rightarrow E(X) = 85/13$$

Q. A continuous Random variable X has density function given by

$$f(x) = \begin{cases} 2e^{-2x} & x > 0 \\ 0 & x < 0 \end{cases}$$

find expected value & variance of X .

$$\begin{aligned} E(X) &= \int_{-\infty}^{\infty} x f(x) dx \\ &= \int_0^{\infty} x (2e^{-2x}) dx. \end{aligned} \quad \left[\begin{aligned} &\int_0^{\infty} x^{n-1} e^{-ax} dx \\ &= \frac{\Gamma(n)}{a^n} \end{aligned} \right]$$

$$= 2 \frac{\Gamma(2)}{2^2} = \frac{1}{2}$$

$$\begin{aligned} \text{Var}(X) &= \int_{-\infty}^{\infty} x^2 f(x) dx = E(X^2) \\ &= \int_0^{\infty} x^2 (2 \cdot e^{-2x}) dx \end{aligned}$$

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$$E(X^2) = \int_0^{\infty} x^2 (2e^{-2x}) dx$$

$$= 2 \int_0^{\infty} x^2 e^{-2x} dx$$

$$= 2 \frac{\sqrt[3]{3}}{2} = \frac{2(2 \cdot 1)}{8} = \frac{1}{2}$$

$$\text{var}(X) = E(X^2) - E(X)^2$$

$$\text{var}(X) = \frac{1}{2} - \frac{1}{4} = \frac{1}{4}$$

$$B(n, p)$$

Binomial Distribution:-

A binomial experiment has following properties :-

① Experiment consist of n -identical and independent trials

② Each trial results in two outcomes : success or failure.

③ The random variable X , is the number of successes in n -trials

④ Probability ^{of success} for all trials is same

$$P(\text{success}) = p$$

$$P(\text{failure}) = q = 1-p$$

ASSUMPTIONS

Examples where Binomial distribution do not hold

eg A box contain 4 red apples, 4 green apples. Select 4 apples without replacement

$X = \text{R.V} = \text{no. of red apples.}$

- Without replacement $\Rightarrow$ dependence

$\rightarrow$ Here the probability of red apples in each trial is not same

Q Toss a coin ^{three} ~~three~~ time. $= 2^3$
Find the prob. of getting
no. of head 1 time. Make P.M.F.

$X \rightarrow$ No. of Head 1 time

$\{ \text{HHT, HTH, THH, HTT, TTH, THT, TTT, HHH} \} = S$

$$P(X) = 3/8$$

OR

By Binomial distribution

$$P(X=1) = {}^3C_1 \left(\frac{1}{2}\right)^1 \left(\frac{1}{2}\right)^2 = 3/8$$

* Repeat an experiment

n , no. of times & n is finite

Probability density ^{Mass} f^n : (P.M.F)

$$p(x) = \begin{cases} {}^nC_x p^x q^{n-x} & ; x=0, 1, 2, \dots, n \\ 0 & ; \text{else} \end{cases}$$

$$p+q=1$$

∴ Random variable X has binomial distribution with parameters n and p .

• Mean = expected value = $np = \mu$

• Variance = $\sigma^2 = npq$

Relation between Mean and variance

$$\frac{\text{Mean}}{\text{Variance}} = \frac{np}{npq} = \frac{1}{q} > 1$$

$$\Rightarrow \frac{\text{Mean}}{\text{Var}} > 1 \Rightarrow \boxed{\text{Mean} > \text{Variance}}$$

Ques If a random variable X has mean 7 & var. 17 then can it be Binomial distribution

Ans No ($\because$ Mean > Var)

Q The probability that man aged 60 will live upto 70 is 0.65, Out of 10 men

Find probability that

- (a) at least 7 will live upto 70
- (b) exactly 9 will live upto 70
- (c) At most 9 will live upto 70

$X =$ No. of Man aged 60 will live upto 70 yrs
 $n = 10$, $p = 0.65$

$$q = 1 - p = 0.35$$

$$(d) P(X) = {}^nC_x p^x q^{n-x}$$

$$P(x) = {}^{10}C_x (0.65)^x (0.35)^{10-x}$$

$$P(X \geq 7) = P(7) + P(8) + P(9) + P(10)$$

$$= {}^{10}C_7 (0.65)^7 (0.35)^3 + {}^{10}C_8 (0.65)^8 (0.35)^2 + {}^{10}C_9 (0.65)^9 (0.35)^1$$

$$+ {}^{10}C_{10} (0.65)^0 (0.35)^{10}$$

$$\Rightarrow P(X \geq 7) = 0.5139$$

$$\textcircled{b} P(9) = {}^{10}C_9 (0.65)^9 (0.35)^1$$

$$P(9) = 0.0725$$

$$\textcircled{c} P(X \leq 9) = 1 - P(X = 10)$$

$$= 1 - P(X = 10)$$

$$= 1 - {}^{10}C_{10} (0.65)^{10} (0.35)^0$$

$$= 0.9865$$

Q Out of 800 families with 5 children each how many families would be expected to have

(i) 3 boys

(ii) 5 girls

(iii) Either 2 or 3 boys

(iv) at least 2 girls

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$$N = 800$$

$$n = 5$$

Let $X = \text{no. of boys}$.

$$p = \frac{1}{2}$$

$$q = \frac{1}{2}$$

$$P(X=x) = {}^nC_x p^x q^{n-x}$$

$$= {}^5C_x \left(\frac{1}{2}\right)^x \left(\frac{1}{2}\right)^{5-x}$$

$$\Rightarrow P(X=x) = {}^5C_x \left(\frac{1}{2}\right)^5$$

$$= {}^5C_x \left(\frac{1}{32}\right)$$

$$P(X=3) = {}^5C_3 \left(\frac{1}{32}\right) = 5/16$$

No. of family have 3 boys

$$= 800 \times \frac{5}{16} = 250$$

$$(ii) P(X=0) = P(5 \text{ girls})$$

$$= {}^5C_0 \left(\frac{1}{32}\right) = \frac{1}{32}$$

No. of families have 5 girls

$$= 800 \times \frac{1}{32}$$

$$= 25$$

$$(iii) P(2) + P(3)$$

$$= {}^5C_2 \left(\frac{1}{32}\right) + {}^5C_3 \left(\frac{1}{32}\right)$$

$$= \frac{20}{32}$$

No. of families have 2 or 3 boys = $800 \times 20/32$

$$(iv) P(\text{at least 2 girls})$$

$$= P(2G) + P(3G) + P(4G) + P(5G)$$

$$= P(3B) + P(2B) + P(1B) + P(0B)$$

$$= {}^5C_3 \left(\frac{1}{32}\right) + {}^5C_2 \left(\frac{1}{32}\right) + {}^5C_1 \left(\frac{1}{32}\right)$$

$$+ {}^5C_0 \left(\frac{1}{32}\right)$$

$$= 0.5$$

No. of families have at least

$$2 \text{ girls} = 800 \times 0.5 = 400$$

Ques for Mean and Variance

Q. A box contains 100 cell phones, 20 of which are defective. 10 cell phones are selected for inspection. What is expected no. ($E(X)$) of defective cells

Soln

$$n = 10$$

$$\begin{aligned} \text{Prob. of defective cells} &= \frac{20}{100} \\ &= \frac{1}{5} \end{aligned}$$

$$\therefore E(X) = \text{expected no. of defective cells} = np = \frac{1}{5} \times 10 = 2$$

Q. For a binomial distribution, mean is 6 and standard deviation is $\sqrt{2}$. Find first two terms of distribution.

Soln

$$\text{Mean} = 6 = np$$

$$S.D = \sqrt{2} \Rightarrow \text{var} = 2 = npq$$

$$(\because S.D = \sqrt{\text{var}})$$

$$\Rightarrow \frac{npq}{np} = q = \frac{2}{6} = \frac{1}{3}$$

$$\therefore p = 1 - q = 1 - \frac{1}{3} = \frac{2}{3}$$

$$\therefore \boxed{p = \frac{2}{3}} \quad \boxed{q = \frac{1}{3}}$$

$$np = 6 \Rightarrow n = \frac{6}{p}$$

$$\Rightarrow n = \frac{6 \times 3}{2} = 9 \Rightarrow \boxed{n = 9}$$

$$P(X=0) = {}^9C_0 \left(\frac{2}{3}\right)^0 \left(\frac{1}{3}\right)^9$$

$$\left(\therefore P(X=x) = {}^nC_x p^x q^{n-x} \right)$$

$$P(X=1) = {}^9C_1 \left(\frac{2}{3}\right)^1 \left(\frac{1}{3}\right)^8$$

- Q A fair die is thrown and an outcome of 4 or 5 is success. If the die is thrown 9 times and X denotes no. of success. Find
- (i) Mean and Variance of X
 - (ii) $P(X=2)$
 - (iii) $P(2X-3=2)$
 - (iv) $P(X \leq 2)$

Soln Let $X =$ outcome of 4 or 5

$$p = \frac{1}{6} + \frac{1}{6} = \frac{2}{6} = \frac{1}{3}$$

$$q = 1 - p = 1 - \frac{1}{3} = \frac{2}{3}$$

$$n = 9$$

$$(i) \text{ mean} = np \\ = 9 \times \frac{1}{3} = 3$$

$$\boxed{\text{mean} = 3}$$

$$\text{var} = npq = 9 \times \frac{1}{3} \times \frac{2}{3} = 2$$

$$\boxed{\text{var} = 2}$$

$$(ii) P(X=2) = {}^9C_2 \left(\frac{1}{3}\right)^2 \left(\frac{2}{3}\right)^7$$

$$(iii) P(2X-3=2) = P(2X=5)$$

$$\Rightarrow P(2X-3) = P(X=2.5) \\ = 0$$

$$\left[\begin{array}{l} \therefore P(X=x) = {}^nC_x p^x q^{n-x} ; x=0,1,\dots,n \\ \quad \quad \quad 0 ; \text{ else} \end{array} \right.$$

$$(iv) P(X \leq 2) = P(X=0) + P(X=1) + P(X=2)$$

$$= {}^9C_0 \left(\frac{1}{3}\right)^0 \left(\frac{2}{3}\right)^9 + {}^9C_1 \left(\frac{1}{3}\right)^1 \left(\frac{2}{3}\right)^8 \\ + {}^9C_2 \left(\frac{1}{3}\right)^2 \left(\frac{2}{3}\right)^7$$

$$P(X \leq 2) = \frac{1 + 9 \times 2^8 + 36 \times 2^7}{3^9}$$

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Q If X follows Binomial distribution with mean 2 and variance $\frac{4}{3}$. Find $P(X=5)$

Soln mean = $np = 2$

$$\text{Variance} = npq = \frac{4}{3}$$

$$\frac{npq}{np} = \frac{4/3}{2} \Rightarrow \boxed{q = 2/3}$$

$$p = 1 - q = 1 - 2/3 \Rightarrow \boxed{p = 1/3}$$

$$np = 2 \Rightarrow \boxed{n = 6}$$

$$\therefore P(X=5) = {}^6C_5 \left(\frac{1}{3}\right)^5 \left(\frac{2}{3}\right)^1$$

$$P(X=5) = \frac{12}{3^6}$$

Q Find M.G.F. of Binomial distribution & find its Mean and variance.

Ans P.M.F: $p(x) = {}^nC_x p^x q^{n-x}$
 $x = 0, 1, 2, \dots$

M.G.F: $M_x(t) = E[e^{tx}]$

$$= \sum e^{tx} \cdot p(x)$$

$$M_x(t) = \sum_{x=0}^n e^{tx} {}^nC_x p^x q^{n-x}$$

$$= \sum_{x=0}^n {}^nC_x (e^t p)^x q^{n-x}$$

$$M_x(t) = (e^t p + q)^n, \quad |e^t p| < q$$

$$\left((a+x)^n = {}^nC_0 a^n x^0 + {}^nC_1 a^{n-1} x^1 + {}^nC_2 a^{n-2} x^2 + \dots + {}^nC_n a^0 x^n \right)$$

; $|x| < a$

Now

$$E[X] = \frac{d}{dt} [MGF]_{t=0}$$

$$E[X] = \frac{d}{dt} [e^t p + q]$$

$$\Rightarrow E[X] = \frac{d}{dt} [e^{tp+q}]^n$$

$$\Rightarrow E[X] = n[e^{tp+q}]^{n-1} p e^t$$

$$\Rightarrow E[X] = n(b+q)^{n-1} p$$

$$\Rightarrow E(X) = np$$

$$E[X^2] = \frac{d^2}{dt^2} [M_X(t)]$$

$$E[X^2] = \frac{d}{dt} [np(e^{tp+q})^{n-1} e^t]$$

$$\Rightarrow E[X^2] = np \left[(n-1)(e^{tp+q})^{n-2} (p e^t) e^t + (e^{tp+q})^{n-1} e^t \right]$$

$$t=0$$

$$\Rightarrow E[X^2] = np \left[(n-1)p + 1 \right]$$

Now

$$\text{var}(X) = E[X^2] - (E[X])^2$$

$$\Rightarrow \text{var}(X) = np[(n-1)p + 1] - n^2 p^2$$

$$\Rightarrow \text{var}(X) = np(1-p)$$

$$\Rightarrow \boxed{\text{var}(X) = npq}$$

How to Fit a Binomial $B(n, p)$ Distribution

<u>Q</u>	$x(i)$ <u>x</u>	<u>f</u>	<u>fx</u>	$n=4$
	0	28	0	$x=0,1,2,3,4$
	1	62	62	
	2	46	92	$P(x) = \text{P.M.F} = {}^nC_x p^x q^{n-x}$
	3	10	30	
	4	4	16	Mean = np
$n=4$ $N=150$			$\begin{matrix} 200 \\ \downarrow \\ \Sigma fx \end{matrix}$	

$$\text{Mean} = \bar{x} = \frac{\Sigma fx}{N} = \frac{200}{150} = \frac{4}{3}$$

For a Binomial Distribution

$$\text{Mean} = np$$

$$\Rightarrow \frac{4}{3} = np \Rightarrow \frac{4}{3} = 4p$$

$$\Rightarrow p = \frac{1}{3} \Rightarrow p = \frac{1}{3} \text{ \& } q = 1-p$$
$$q = \frac{2}{3} \quad n=4$$

$$P(X=x) = {}^nC_x p^x q^{n-x}$$

$$P(0) = {}^4C_0 \left(\frac{1}{3}\right)^0 \left(\frac{2}{3}\right)^4 = \frac{16}{81}$$

$$P(1) = {}^4C_1 \left(\frac{1}{3}\right)^1 \left(\frac{2}{3}\right)^3 = \frac{32}{81}$$

$$P(2) = {}^4C_2 \left(\frac{1}{3}\right)^2 \left(\frac{2}{3}\right)^2 = \frac{24}{81}$$

$$P(3) = {}^4C_3 \left(\frac{1}{3}\right)^3 \left(\frac{2}{3}\right) = \frac{8}{81}$$

Teleo $\frac{DL:}{Pg:}$

$$P(4) = {}^4C_4 \left(\frac{1}{3}\right)^4 \left(\frac{2}{3}\right)^0 = \frac{1}{81}$$

Binomial Distribution

x	$P(x)$	$N \times P(x)$	(Ei) Expected frequency	(Oi) Observed Frequency
0	$16/81$	29.63	≈ 30	28
1	$32/81$	59.26	≈ 59	62
2	$24/81$	44.44	≈ 44	46
3	$8/81$	14.81	≈ 15	10
4	$1/81$	1.85	≈ 2	4
$\Sigma f = N = 150$				<u>150</u>

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① Poisson Distribution

Let $\lambda > 0$ be fixed. The counting process $[N(t), t \in [0, \infty)]$ is called Poisson Process with rate λ if

- ① $N(0) = 0$
- ② $N(t)$ has independent incremental
- ③ The number of arrivals in any interval of length $t > 0$ has Poisson Distribution

• Poisson Distribution usually used in case when we are counting the occurrences of certain events that appear to happen at certain rate but completely random.

Examples

- ① No. of snake bites in a city
- ② No. of car accidents at a site
- ③ No. of phone calls at a booth.

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Poisson Distribution

Analysis :- Binomial Distribution is used when n is finite & probability of success is approximately $1/2$

Poisson Distribution is used when n is very large & p (success) is very small.

Poisson Distribution :

Defn :- A discrete random variable X which has following Probability Mass function as $p(x) = \frac{\lambda^x e^{-\lambda}}{x!}$, $x=0, 1, 2, \dots$

is called Poisson Variate and its distribution is called Probability Distribution of Poisson

Imp Prove that Poisson Distribution is limiting case of Binomial Distribution under following conditions :

- | | |
|----------------------------|---|
| (i) $n \rightarrow \infty$ | $\left\{ \begin{array}{l} \text{NOTE: } n \text{ is very large} \\ \text{in comparison of } p \end{array} \right\}$ |
| (ii) $p \rightarrow 0$ | |
| (iii) $np = \lambda$ | |

Proof $P(X=x) = {}^nC_x p^x q^{n-x}$

$$P(X=x) = \frac{n!}{(n-x)! x!} p^x q^{n-x}$$

Good Write

(3)

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$$\text{Now } \lim_{n \rightarrow \infty} P(X=x)$$

$$= \lim_{n \rightarrow \infty} \frac{n!}{(n-x)! x!} p^x (1-p)^{n-x}$$

$$= \lim_{n \rightarrow \infty} \frac{n(n-1)(n-2) \dots (n-(x-1)) \cancel{(n-x)!}}{x! \cancel{(n-x)!}} p^x (1-p)^{n-x}$$

$$= \left(\because np = \lambda \Rightarrow n = \lambda/p \right) \\ \text{or } p = \lambda/n$$

$$= \lim_{n \rightarrow \infty} \left[\frac{n^x \left[\left(1 - \frac{1}{n}\right) \left(1 - \frac{2}{n}\right) \dots \left(1 - \frac{x-1}{n}\right) \right] \left(\frac{\lambda}{n}\right)^x \left[\frac{1-\lambda}{n}\right]^{n-x}}{x!} \right]$$

$$= \lim_{n \rightarrow \infty} \frac{\lambda^x}{x!} \left(1 - \frac{\lambda}{n}\right)^n \left(1 - \frac{\lambda}{n}\right)^{-x}$$

$$= \frac{\lambda^x}{x!} \lim_{n \rightarrow \infty} e^{-\lambda}$$

$$\because \lim_{x \rightarrow \infty} \left(1 + \frac{a}{x}\right)^x = e^a$$

$$\lim_{n \rightarrow \infty} P(X=x) = \frac{\lambda^x}{x!} e^{-\lambda}$$

→ Poisson distribution function

$$P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!}, \quad x=0, 1, 2, 3, \dots$$

Mean = variance = λ

$$\text{Mean} = \lambda$$

$$\text{Variance} = \lambda$$

Q Between the hours 2 P.M. and 4 P.M. the average no. of phone calls per minute coming into the cellphone company is 2.35. Find the probability that during one particular minute, there will be at most 2 phone calls.

So Mean = 2.35 $\lambda = 2.35$

$$P(X \leq 2) = P(X=0) + P(X=1) + P(X=2)$$

$$= \frac{e^{-2.35} (2.35)^0}{0!} + \frac{e^{-2.35} (2.35)^1}{1!} + \frac{e^{-2.35} (2.35)^2}{2!}$$

$$P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

[$\therefore$ Between 2 P.M. and 4 P.M., average no. of calls is given. But total no. of calls (n) is not given so $n \rightarrow \infty$]

Q. It is known that past experience that in a certain plant there are on average 4 industrial accidents per month. Find the probability that in a given year there will be less than 4 accidents.
 $\lambda = 4$

Soln $P(X < 4) = P(X=0) + P(X=1) + P(X=2) + P(X=3)$

$$\Rightarrow P(X < 4) = \frac{e^{-4} 4^0}{0!} + \frac{e^{-4} 4^1}{1!} + \frac{e^{-4} 4^2}{2!} + \frac{e^{-4} 4^3}{3!}$$

$$P(X < 4) = e^{-4} \left(1 + 4 + 8 + \frac{32}{3} \right)$$

Q. Fit the Poisson distribution to following data and calculate the theoretical frequencies

x	0	1	2	3	4
f	123	59	14	3	1

Soln To get poisson distribution, we need parameter λ

Actual
frequencies

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x	f	fx
0	123	0
1	59	59
2	14	28
3	3	9
4	1	4
Total	200	100

$$\text{Mean} = \frac{\sum fx}{\sum f}$$

$$\text{Mean} = \frac{100}{200} = 0.5$$

$$\Rightarrow \lambda = 0.5$$

Theoretical freq.
 $200 \times \frac{e^{-\lambda} \lambda^x}{x!}$

$$P(X=x) = \frac{e^{-\lambda} \lambda^x}{x!}$$

121.3	~	121
60.65		61
15.3125		15
2.552		3
0.32		0

200

Total of Actual frequencies = Theoretical frequencies

Q If 5% of the electric bulb manufactured by a company are defective, find the probability that in a sample of 100 Bulbs

- none is defective
- 5 bulbs will be defective

Good Write

(i) Required Probability = $P(X=0)$

~~$p = \lambda = 0.05$~~

$$P(X=0) = \frac{e^{-\lambda} \lambda^x}{x!}$$

$$\Rightarrow P(X=0) = \frac{e^{-5} (5)^0}{0!} = e^{-5}$$

$$\boxed{P(X=0) = e^{-5}}$$

$$[\because \lambda = np = 100 \times \frac{5}{100} = 5]$$

$$(ii) P(X=5) = \frac{e^{-5} (5)^5}{5!}$$

Q Find M.G.F. of Poisson Distribution. Does $M_x(t)$ exist for all values of t

$$p(x) = \frac{e^{-\lambda} \lambda^x}{x!} ; x=0,1,2,\dots$$

M.G.F. = $E[e^{tx}]$
M.G.F. is $M_x(t) = \sum_{x=0}^{\infty} e^{tx} \left(\frac{e^{-\lambda} \lambda^x}{x!} \right)$

$$M_x(t) = e^{-\lambda} \sum_{x=0}^{\infty} \frac{(e^t \lambda)^x}{x!}$$

$$\Rightarrow M_x(t) = \frac{e^{-\lambda}}{e^{-\lambda}} e^{e^t \lambda} = e^{-\lambda} e^{e^t \lambda}$$

$\because e^x = \sum_{n=0}^{\infty} \frac{x^n}{n!}$ and is convergent $\forall x$

$$\therefore M_x(t) = e^{-\lambda} (1 - e^t)$$

which exist $\forall t$

Q Find Mean and Variance from M.G.F. in Poisson distribution

$$E[X] = \frac{d}{dt} [e^{-\lambda(1-e^t)}]_{t=0}$$

$$E[X] = e^{-\lambda(1-e^t)} (\lambda e^t)$$

$$\Rightarrow E[X] = e^0 \lambda e^0 = \boxed{\lambda = \mu = \text{mean}}$$

$$\text{Now } E(X^2) = \frac{d^2}{dt^2} [e^{-\lambda(1-e^t)}]$$

$$= \frac{d}{dt} [e^{-\lambda(1-e^t)} (\lambda e^t)]$$

$$E[X^2] = e^{-\lambda(1-e^t)} (\lambda e^t) + e^{-\lambda(1-e^t)} (\lambda e^t)^2$$

$$\Rightarrow E[X^2] = e^0 \lambda e^0 + e^0 (\lambda e^0)^2 = \lambda + \lambda^2$$

$$\text{Var}(X) = E(X^2) - (E(X))^2 = \lambda + \lambda^2 - \lambda^2$$

$$\boxed{\text{Var}(X) = \lambda}$$

Continuous Distribution

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Normal Distribution

probability density function =

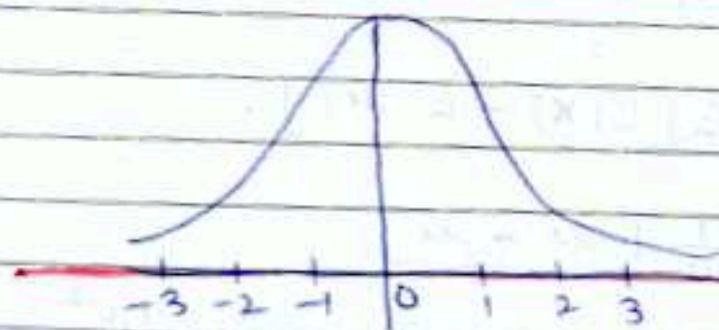
$$p.d.f = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}; -\infty < x < \infty$$

$$\text{Mean} = E(X) = \mu$$

$$\text{Variance} = \text{Var}(X) = \sigma^2$$

$$\text{Notation } X \sim N(\mu, \sigma^2)$$

Also known as Gaussian Distribution
or Bell shaped curve Distribution



Properties:-

- ① Mean = Median = Mode
- ② Curve is symmetric about mean
- ③ Total area under curve is 1 $\rightarrow \because p.d.f = \int_{-\infty}^{\infty} f(x) dx = 1$
- ④ Left area and right area about mean are same i.e. 0.5

For Normal distributed variable

$$X \sim N(\mu, \sigma^2)$$

$$Z = \frac{X - \mu}{\sigma}$$

$Z \rightarrow$ called Standard Normal variate

$$\bullet \text{ P.d.f} = f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}$$

$\bullet$ Mean of Z

$$\begin{aligned} E(Z) &= E\left(\frac{X - \mu}{\sigma}\right) \left[\begin{array}{l} \because E[aX] \\ = aE[X] \\ E[X+Y] = E[X] + E[Y] \end{array} \right] \\ &= \frac{1}{\sigma} E(X - \mu) \\ &= \frac{1}{\sigma} [E(X) - E(\mu)] \\ &= \frac{1}{\sigma} [\mu - \mu] \end{aligned}$$

$$E(Z) = 0$$

$$Z \sim N(0, 1)$$

$$\text{var}(Z) = \text{var}\left(\frac{X - \mu}{\sigma}\right)$$

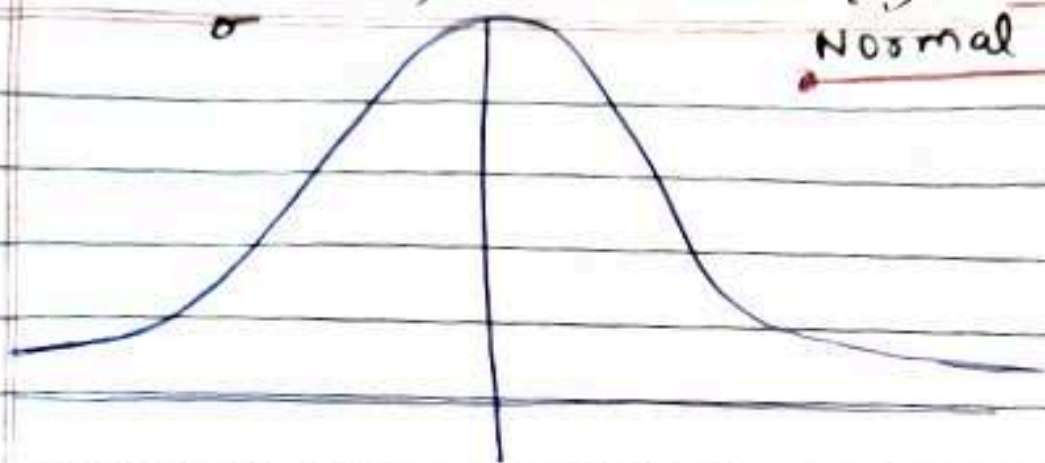
$$= \frac{1}{\sigma^2} \text{var}(X - \mu) \quad (\text{var}(\text{constant}) = 0)$$

$$= \frac{1}{\sigma^2} [\text{var}(X) + \text{var}(\mu)]$$

$$= \frac{1}{\sigma^2} (\sigma^2) = 1 = \text{var}(Z)$$

• $X \sim N(\mu, \sigma^2) \rightarrow$ Normal Distribution

• $Z = \frac{X - \mu}{\sigma} \Rightarrow Z \sim N(0, 1) \rightarrow$ Standard Normal Distribution



• STANDARD NORMAL DISTRIBUTION

• Pd.f = $\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2}$

• C.D.F = $\Phi(z) = P(Z \leq z) = \int_{-\infty}^z \phi(z) dz$

Q. If X is Normal variate with mean = 30 & standard deviation = 5. then find probability that

(i) $26 \leq X \leq 40$

(ii) $X \geq 45$

(iii) $|X - 30| > 5$

Soln ~~206~~ $\mu = 30$ $\sigma = 5$

$$Z = \frac{X - \mu}{\sigma} \Rightarrow Z = \frac{X - 30}{5}$$

(i) $26 \leq X \leq 40$

$$\frac{26 - 30}{5}, \frac{40 - 30}{5}$$

$$\Rightarrow \frac{-4}{5}, \frac{10}{5}$$

$$= -0.8, 2$$

$$\therefore P(26 \leq X \leq 40) = P(-0.8 \leq Z \leq 2)$$

$$= P(-0.8 \leq Z \leq 0) + P(0 \leq Z \leq 2)$$

$$= 0.2881 + 0.4772$$

$$= 0.7653$$

(ii) $X > 45$

$$Z = \frac{45 - 30}{5} = 3$$

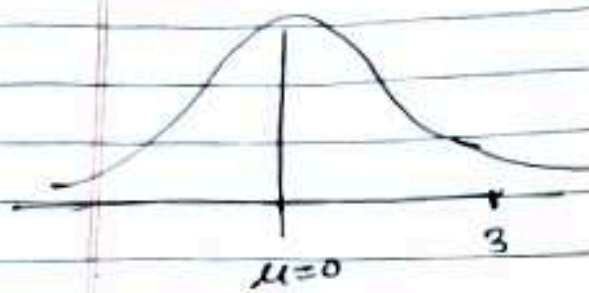
$$P(0 \leq z \leq 3) = P(z=3) - P(z=0) \\ = 0.9987 - 0.5000 = 0.4987$$

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$$P(X \geq 45) = P(Z \geq 3) = P(Z > 0) - P(0 \leq z \leq 3)$$

$$= 0.5 - 0.4987$$

$$= 0.0013$$



$$(III) |X - 30| > 5$$

$$|X - 30| \leq 5$$

$$X - 30 \leq 5$$

$$X \leq 35$$

$$-(X - 30) \leq +5$$

$$-X \leq -25$$

$$X \geq 25$$

$$\text{Now } P(|X - 30| > 5) = 1 - P(|X - 30| \leq 5) \\ = 1 - P(25 \leq X \leq 35)$$

$$= 1 - P(-1 \leq Z \leq 1)$$

$$= 1 - 2P(0 \leq Z < 1)$$

$$= 1 - 2 \times 0.3413$$

$$= 1 - 0.6826$$

$$P(|X - 30| > 5)$$

$$= 0.3174$$

$$P(-0.8 \leq z \leq 0) = P(z < 0) - P(z < -0.8) \\ = 0.5 - 0.2119 = 0.2881$$

$$P(0 < z < 2) = P(z > 0) - P(z > 2) \\ = 0.5 - P(z < -2)$$

$$= 0.5 - 0.0228$$

Good Write

$$= 0.4772$$

Q In a Normal Distribution of marks, 31% are under 45 & 8% are over 64. Find Mean & standard deviation of the distribution. It is given that if

$$f(t) = \frac{1}{\sqrt{2\pi}\sigma} \int_0^t e^{-x^2/2} dx \quad \text{then}$$

$$f(0.5) = 0.19 \quad \& \quad f(1.4) = 0.42.$$

Q Let μ & σ denote mean & S.D. respectively. $\therefore$ 31% are under 45, the area to the left of ordinate at $x=45$ is 0.31.

Also, since 8% of marks are over 64, the area to the right of ordinate at $x=64$ is 0.08.

$$x=45, \quad z_1 = \frac{45-\mu}{\sigma} \quad (z_1 = ?)$$

$$P(-\infty < x < 45) = 0.31$$

$$\Rightarrow P(-\infty < z < z_1) = 0.31$$

$$\Rightarrow P(-\infty < z < 0) - P(z_1 < z < 0) = 0.31$$

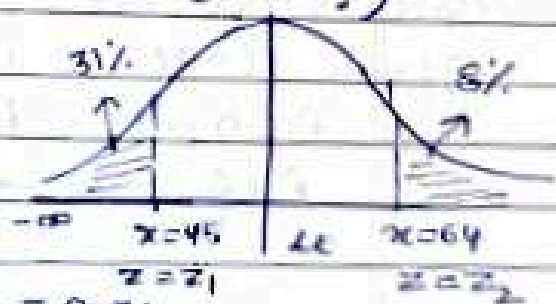
$$\Rightarrow 0.5 - P(z_1 < z < 0) = 0.31$$

$$\Rightarrow 0.5 - 0.31 = P(z_1 < z < 0) \Rightarrow P(z_1 < z < 0) = 0.19$$

$$\text{Also } f(0.5) = 0.19$$

$$\Rightarrow P(z_1 < z < 0) = P(z_1) = f(0.5) = 0.19$$

$$\Rightarrow \boxed{z_1 = 0.5} \quad \text{But on L.H.S } z_1 = -0.5$$



$$x=64 \quad P(Z_2 < Z < \infty) = 0.08$$

$$\Rightarrow P(0 < Z < \infty) - P(0 < Z < Z_2) = 0.08$$

$$\Rightarrow 0.5 - P(Z_2) = 0.08$$

$$\Rightarrow P(Z_2) = 0.5 - 0.08 = 0.42$$

$$f(1.4) = f(Z_2) = 0.42$$

$$\boxed{Z_2 = 1.4}$$

$$\therefore Z_1 = \frac{x - \mu}{\sigma} \quad Z_2 = \frac{x - \mu}{\sigma}$$

$$0.5 = \frac{45 - \mu}{\sigma}$$

$$1.4 = \frac{64 - \mu}{\sigma}$$

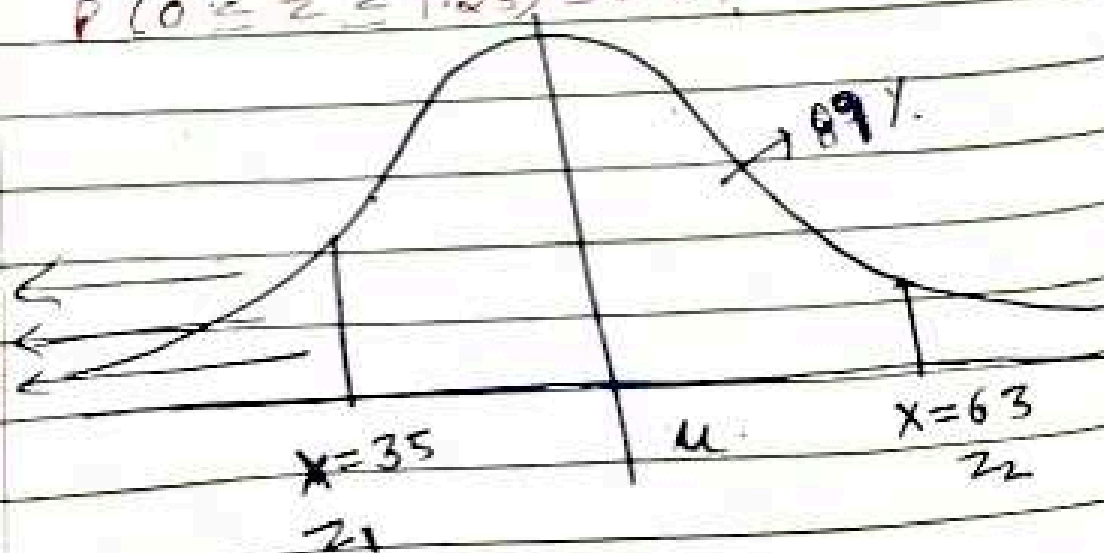
$$\Rightarrow \left. \begin{array}{l} 45 - \mu = -0.5\sigma \\ 64 - \mu = 1.4\sigma \end{array} \right\} \text{ solving } \Rightarrow \sigma = 10, \mu = 50$$

Q In a normal distribution, 7% of items are under 35 & 89% are under 63. Find mean and variance of the distribution given that

$$P(0 \leq Z \leq 0.18) = 0.07$$

$$P(0 \leq Z \leq 1.48) = 0.43$$

$$P(0 \leq Z \leq 1.23) = 0.39$$



$\therefore 7\%$ of items are under 35

$\therefore 50 - 7 = 43\%$ " " lie b/w μ & 35

Now value

$$\therefore P(0 \leq Z \leq 1.48) = 0.43 \quad \text{given}$$

$$\therefore P(0 \leq Z \leq z_1) = 0.43$$

$$\Rightarrow z_1 = 1.48 \quad \Rightarrow \boxed{z_1 = -1.48} \quad (\text{on L.H.S})$$

$$z_1 = \frac{x - \mu}{\sigma} \Rightarrow -1.48 = \frac{35 - \mu}{\sigma} \quad \text{--- (1)}$$

$\therefore 89\%$ are under 63

$$P(x_1 < x < x_2) =$$

$$\Rightarrow 89\% - 50\% = 39\% \text{ lie b/w } \mu \text{ \& } x = 63$$

$$\Rightarrow P(\mu < x < 63) = 39\%$$

$$\Rightarrow P(0 < Z < z_2) \approx 0.39 \approx P(0 \leq Z \leq 1.23)$$

$$\Rightarrow \boxed{z_2 = 1.23} \quad \Rightarrow 1.23 = \frac{63 - \mu}{\sigma} \quad \text{--- (2)}$$

Solve (1) & (2) $\Rightarrow \mu = 50.3$
 $\sigma = 10.33$